

John Robison Hoopes

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London, UK

Researcher and engineer working on verifiable geospatial systems and their application to AI governance and safety.

SUMMARY

I'm a researcher, developer, and founder, with a through-line in **verifiable geospatial systems**. I've spent a decade applying geospatial data technologies for impact, in maritime security and arms control, carbon markets, and national location data.

I'm now focused on **verifiable AI governance**. Location and geospatial verification provide a technical foundation for enforceable agreements about where AI systems run and how intelligent machines behave.

EXPERIENCE

University of Maryland — Research Affiliate

2025 – present

Verifiable location and spatial data integrity, Dept. of Geographical Sciences.

- Researching **location verification** and spatial data integrity — composable location evidence, and credibility evaluation for claims from untrusted sources. With **Prof. Taylor Oshan**.
- Designed and developed the **Location Protocol** — an open standard for sharing location data and proofs across heterogeneous, decentralized systems. Supported by grants from the **National Science Foundation** and the **Filecoin Foundation for the Decentralized Web**.
- Contributing to **Open Geospatial Consortium** standards — the Blockchain & Distributed Ledger Technologies Domain Working Group, and the Integrity, Provenance & Trust effort (Testbed-21).
- Speaker: OGC Members Meeting, FOSS4G North America 2025, Devconnect Buenos Aires; published at GISRUUK 2025.

Astral — Founder & Lead Developer

2019 – present

Verifiable location infrastructure for apps, agents, and smart contracts.

- Designed and built an open source protocol providing verifiable location-based services.
- **Multi-factor location proofs** — composing independent signals into hard-to-forge evidence of the location and timing of an event, person, or device. A new class of spatial trust primitives.
- Geospatial **policy evaluation** inside **trusted execution environments**; attestation pipelines; support for **privacy-preserving proofs**, so a device can demonstrate location-based policy compliance without revealing its position.

Toucan — Co-Founder & Head of Ecosystem

2021 – 2023

Market infrastructure for carbon credits, operating at scale.

- Co-founded venture-backed infrastructure that managed carbon credits with **smart contracts** — >\$4B in market activity settled, ~4% of Verra's circulating supply tokenized at launch.
- Built and led a **30-person team**; raised from **USV** and others. Coordinated across engineering, enterprise buyers, standards bodies, and policymakers.

Ordnance Survey — Developer Advocate + Innovation Research

2020 – 2021

Developer advocacy and applied research at Great Britain's national mapping agency.

- Led the developer ecosystem for the **OS Data Hub** launch — Britain's national location data, opened via API under a **£1B government licensing agreement**.
- Innovation research on verifiable geospatial systems: **smart contracts**, **decentralized identifiers (DIDs)**, and **content-addressed storage** for geospatial data.

ADDITIONAL ROLES

- Machine Learning Institute** — Machine Learning Fellow. Intensive fellowship building ML systems end-to-end: training pipelines, embeddings, vector search, transformers, multimodal architectures, containerized deployment 2024, 2026
- Shuraako / One Earth Future** — Geospatial Data Product Engineer & Research Assistant. 2016 – 2018
Data science for the **State of Maritime Piracy**; built the **Stable Seas** data visualization (reached UN, IMO); researched location spoofing for maritime sanctions evasion.
- London Blockchain Labs** — Technical Director. Taught postgraduate cryptography, CS, and data science; teams won multiple hackathons. 2019 – 2020
- World Economic Forum** — Co-Chair, Web3 Climate Action Working Group; lead technical co-author of the working-group white paper. 2023
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SKILLS

- Verifiable Systems** Location verification, multi-factor proofs, secure hardware / trusted execution environments, attestation, smart contracts, protocol & standards design (OGC). Privacy-preserving proofs — zero-knowledge, selective disclosure.
- Geospatial** Spatial data science, spatial data pipelines at scale, PostGIS, QGIS, Mapbox GL, verifiable geocomputation.
- Engineering** Python, TypeScript, Solidity, SQL, Docker, REST APIs, cloud deployment, ML systems.
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EDUCATION

- UCL** — MSc Spatial Data Science & Visualisation, *Distinction*. 2018 – 2019
Dissertation: *Sovereign Sensors* — secure-hardware IoT sensors writing to smart contracts; verifiable sensor networks on distributed ledgers.
- Tufts University** — BSc Biology & Environmental Studies, *Neubauer Scholar*. 2006 – 2010
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SELECTED PUBLICATIONS

- Spatial Alignment — *SOTA Letters*, 2026
- Towards Harder Location Proofs — *Flashbots Collective*, 2025
- Towards a Decentralized Geospatial Web — *GISRUUK 2025*, with T. Oshan
- State of Provenance Verification for Remote Sensing Imagery — *johnx.co*, 2026
- Intro to Multifactor Location Verification — *Frontier Forum*, 2025
- Astral: A Spatial Extension for the Decentralized Web — *FOSS4G NA 2025*
- Blockchain for Scaling Climate Action — *World Economic Forum white paper*, 2023
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BEYOND WORK

- Tour Aotearoa** — solo bikepacked ~3,000 km the length of New Zealand (2025).
- S/V Daedalus** — co-own and sail a 29-ft cruising boat around the UK and Europe.
- Casa Bonita cliff diver** (2006) — my best job; I peaked early.